

PAVAN KUMAR KR

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PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with 4+ years of experience building scalable applications, deploying AI/LLM systems, and architecting cloud-based solutions. Strong expertise in LLM inference, RAG pipelines, vector databases, chatbot development, and multi-model orchestration. Proven ability to own end-to-end product development—from backend microservices to infrastructure, CI/CD, and production support. Collaborated directly with leadership to drive technical strategy and deliver business-impactful AI solutions.

SKILLS

AI / LLM / ML

- LLM Inference & Optimization: vLLM, HuggingFace Transformers, TensorRT-LLM.
- RAG (Retrieval-Augmented Generation): Weaviate.
- Prompt Engineering, Fine-tuning, PEFT, LoRA.
- Multi-model orchestration & agents.
- LLM deployment: FastAPI, Nginx.
- NLP Tasks: NL-to-SQL model integration, embeddings, tokenization.
- Model serving: multiple concurrent models / multi-thread agent pipelines
- Programming: Python, Java, C#, JavaScript
- Python libraries: NumPy, Pandas, scikit-learn, TensorFlow
- Frameworks: Django, Odoo, FastAPI, Spring Boot, .NET
- Frontend: React (Vite + TS), HTML, CSS, jQuery, Tailwind, Bootstrap
- Databases: PostgreSQL, MySQL, MongoDB, NoSQL
- Cloud: AWS, Azure, GCP
- DevOps: Nginx, Apache, Jenkins
- Tools: GitHub, Bitbucket, Solr, n8n workflows

WORK HISTORY

Dec 2022 - Present

Full Stack Developer + Principal Engineer

Codeland Infosolutions Pvt. Ltd
Bengaluru, India

- Designed and deployed an AI-powered chatbot with RAG, embeddings, and multi-model routing.
- Implemented LLM inference servers using vLLM, supporting multiple concurrent models for different use-cases (text generation, embeddings, NL-to-SQL).
- Developed backend microservices (Spring Boot + Python) with high scalability, resiliency, and security.
- Built REST APIs integrated with React (Vite + TS) and mobile apps.
- Optimized cloud infrastructure, reducing compute costs by 40%.
- Architected production deployments across AWS, Azure, and GCP.
- Led cross-functional teams and conducted code reviews.
- Designed n8n workflow automations, infrastructure monitoring, and logging.

July 2022 - Dec 2022

Junior Python Developer

Codeland Infosolutions Pvt. Ltd
Bengaluru, India

- Developed Python services and API modules.
- Improved front-end interactivity and user engagement.
- Reduced server costs by 10% through automated resource cleanup and optimization.

PROJECTS HANDLED

Project #1: Codeland AI(2025 – Present) - Product

Description: A complete business automation and tracking platform integrating multiple services into one model. Designed to streamline workflows, enhance productivity, and centralize IT operations.

Tech: Python, Java, React (Vite+TS), vLLM, Weaviate, PostgreSQL, Nginx, n8n.

Key Responsibilities: Architected a full AI automation platform integrating chatbots, workflow automations, and RAG. Designed multi-model serving (LLM + Embeddings + Custom NL-to-SQL model). Built the full stack from infrastructure to frontend. Implemented vector search pipelines using Weaviate. Optimized LLM inference using vLLM for high concurrency and low latency.

Project #2: Collection Automation & Barcoding System (2022 – Present) - Product

Description: The CABS (Collection and Barcoding System) web application acts as a centralized platform for receiving and managing data collected by drivers and pickers who gather BMW waste from HCEs. It facilitates the scanning of barcoded waste bags and presents the collected data in the form of reports for analysis and tracking purposes.

Tech: Python, PostgreSQL, Nginx.

Key Responsibilities: Custom module development, business requirement analysis, database schema management, performance optimization, technical support, SQL query optimization.

Project #3: MyCropMonitoring (2024 – 2025) – IT Service

Description: MyCropMonitoring is an agricultural application designed to help farmers monitor and manage crop cultivation and sustainability. Features include crop activity scheduling, soil health assessment, tree health monitoring, crop economics analysis, bulk sales tracking, and sustainability metrics.

Tech: ASP.NET 4.8, C#, MySQL, Azure.

Key Responsibilities: Developing and maintaining application modules, writing and testing clean code for both front-end and back-end, collaborating with team members to define project requirements, managing MySQL databases for data operations, troubleshooting and debugging to resolve technical issues, ensuring compliance with performance and security standards, and participating in code reviews to maintain quality.

Project 4#: Meta (2022 – 2024) - Product

Description: The project aimed to develop customized, user-friendly solutions for streamlining invoice and payment processes. It included functionalities for managing employee travel expenses, real-time employee tracking, an HR ticketing system, and generating reports on marketing activities conducted through buses and billboards.

Tech: Python 3.6, Odoo Framework, Solr, Javascript.

Key Responsibilities: Collaborating with stakeholders to gather requirements, designing and optimizing database schemas, writing efficient SQL queries, developing custom modules, integrating APIs, generating reports, testing invoice features, and providing technical support.

Project 5#: ITC (2022 – 2024) – IT Service

Description: This web application provides a platform that enables waste generators, waste pickers, and recyclers to sell and buy recyclables. Transporters and waste pickers collect waste from HCEs and scan barcoded waste bags. The application receives the data sent by drivers and presents it in the form of JavaScript reports.

Key Responsibilities: Developing odoo modules required for the web application. Creating API's, Creating Qweb and Javascript based reports and dashboard.,Creating linux based cron jobs for automatic email trigger and sms integration. Analyze diagnose and resolve errors related to the application. Managing the cloud server and trouble shooting the issues.

Project #6: Marketing Executive Payment Collection (2022 – 2023) - Product

Description: The project's goal is to develop a web application for marketing executives to efficiently collect monthly payments from hospitals. The application provides a clear overview of all received payments and pending invoices through detailed reports and dashboards, facilitating easy oversight and action by supervisors.

Key Responsibilities: API development, report generation, invoice module implementation.

EDUCATION

Bachelor of Engineering in Civil | Brindavan College of Engineering | Bangalore | Graduated: Aug 2020

Web Development, Python Programming, AL & ML | Duration: Jan 2022 to July 2022